CRIMEVISIONAI

Problem–Solution Fit

Folder 3 | Fit hypothesis and validation plan | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Problem / solution hypothesis

Problem: reviewing heterogeneous evidence requires switching between tools and manually reconstructing observations. **Solution hypothesis:** a case-centric web interface that runs first-pass audio, image and video analysis and organizes candidate events can reduce review friction while leaving final interpretation to a human reviewer.

2. Fit mapping

User problem	CrimeVisionAI response	Current fit assessment	Validation still needed
Evidence scattered across files	Case records and case-scoped uploads	Prototype capability exists	Walk through a case with mixed modalities.
Speech is hard to search	Whisper transcript with timestamps	Basic capability exists	Measure transcription quality on representative audio.
Visual review is time-consuming	Object detections, CLIP labels, BLIP captions	Basic capability exists	Evaluate precision/recall and failure cases.
Chronology is difficult to assemble	Timestamped candidate event timeline	Basic capability exists	Check time alignment, duplicates and source references.
Questions require manually finding observations	Case-level chat over available analysis	Partial fit; known grounding issue	Use a fixed Q&A test set and score evidence coverage.
Formal reporting is repetitive	Structured PDF report export	Not confirmed complete	Implement and verify export before claiming fit.

3. Differentiation and limitations

The prototype combines several modalities in one workflow rather than relying on a single image or text model. However, integration alone does not establish forensic accuracy. Model outputs can miss objects, miscaption scenes, misrecognize speech or generate unsupported narratives. The system is an investigative aid and must retain human verification.

4. Validation plan

1) Prepare a controlled set of authorized or synthetic audio, images and videos with known contents. 2) Record expected transcript segments, objects, timestamps and relevant entities. 3) Run each pipeline under documented hardware and configuration. 4) Compare output with ground truth. 5) Test timeline ordering and Q&A; grounding, including known failures. 6) Record misses, false positives, latency and user task completion. 7) revise scope based on evidence, not assumptions.

5. Fit conclusion

Preliminary fit is plausible for a student prototype and demonstration. Strong product-market fit, forensic validity, measurable efficiency improvement and production readiness have not been established. These require user evaluation, benchmark datasets, documented tests and stronger evidence provenance.